

SHARYPOV, Russian Federation

WMO: 295580

Lat: 55.5344N

Lon: 89.2053E

Elev: 319

StdP: 97.55

Time Zone: 7.00 (E07)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.2	-31.8	-37.6	0.1	-35.1	-34.4	0.2	-31.7	9.9	-9.4	8.3	-7.9	0.7	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	28.6	18.5	26.9	18.1	25.3	17.3	20.2	26.3	19.3	25.0	18.5	23.6	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.1	13.5	23.0	17.2	12.7	21.9	16.3	12.1	21.2	59.6	26.0	56.7	25.1	53.7	23.5	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.5	5.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.9	-15.6	-13.6	-4.6	3.8	8.7	17.0	18.3	16.3	9.6	1.8	-8.7	-11.6		
	DBStd	13.37	8.19	8.16	6.40	5.36	4.97	4.28	2.77	3.30	4.49	5.34	8.27	8.51		
	HDD10.0	3748	794	661	453	193	88	6	0	1	61	258	562	671		
	HDD18.3	6098	1052	894	711	436	300	72	36	79	264	512	812	929		
	CDD10.0	776	0	0	1	7	47	215	256	197	48	4	0	0		
	CDD18.3	84	0	0	0	0	1	31	33	18	1	0	0	0		
	CDH23.3	859	0	0	0	2	26	370	284	153	25	0	0	0		
	CDH26.7	168	0	0	0	0	1	88	51	24	4	0	0	0		
(14) Wind	WSAvg	2.5	2.5	2.4	2.6	3.0	2.9	2.1	1.9	1.9	2.2	2.6	2.8	2.7		
(15) Precipitation	PrecAvg	538	21	18	24	34	46	70	74	77	53	46	40	33		
	PrecMax	628	44	38	56	73	90	129	117	151	112	78	66	60		
	PrecMin	436	3	5	6	6	17	40	31	27	8	24	21	10		
	PrecStd	57	9	8	12	15	21	21	22	28	23	14	12	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.9	3.8	12.2	22.1	26.0	31.4	30.3	29.1	26.7	18.9	10.2	3.5		
		MCWB	-0.4	0.7	6.4	10.4	13.2	18.9	20.3	20.3	15.2	10.0	6.7	0.9		
	2%	DB	0.4	1.3	8.4	17.4	23.4	29.0	27.9	26.9	22.7	14.7	5.5	1.6		
		MCWB	-1.7	-1.0	3.2	8.3	12.3	17.8	19.4	18.6	14.9	8.4	2.6	-0.3		
	5%	DB	-2.0	-0.2	6.0	15.3	21.0	27.3	26.4	24.9	19.5	11.8	3.0	0.2		
		MCWB	-3.8	-2.1	2.2	7.7	11.2	17.5	18.7	18.0	13.2	7.4	0.6	-1.6		
	10%	DB	-4.1	-2.0	4.0	12.5	17.8	25.5	24.7	23.1	17.1	9.2	1.2	-1.4		
		MCWB	-5.7	-3.7	0.7	6.4	10.3	16.7	18.1	17.3	12.0	5.8	-0.7	-3.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.1	1.1	6.2	11.2	14.9	21.1	21.6	21.4	16.7	10.9	6.4	1.2		
		MCDB	2.0	3.5	11.1	18.7	22.6	28.2	28.7	26.9	23.7	16.8	9.9	2.9		
	2%	WB	-1.9	-0.7	3.9	9.8	13.5	19.8	20.3	19.6	15.1	9.2	2.5	-0.1		
		MCDB	0.4	0.9	7.7	16.0	20.3	25.8	26.3	25.5	20.8	14.0	4.7	1.4		
	5%	WB	-3.9	-2.1	2.3	8.2	12.3	18.8	19.5	18.7	13.9	7.6	0.7	-1.6		
		MCDB	-2.1	-0.4	5.7	13.9	19.3	24.8	25.1	23.6	19.2	11.4	2.6	0.0		
	10%	WB	-5.8	-3.9	0.9	6.6	11.0	17.7	18.6	17.7	12.5	6.0	-0.6	-3.1		
		MCDB	-4.2	-2.1	4.0	12.3	17.7	23.7	23.5	22.1	16.4	9.5	1.3	-1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.8	9.4	10.2	10.8	11.8	10.0	10.0	9.4	7.5	6.9	7.4		
		MCDBR	8.4	8.5	10.5	14.6	16.1	15.0	13.4	13.3	14.3	12.1	7.1	7.2		
	5% WB	MCWBR	7.0	6.8	6.9	7.6	7.7	6.2	6.1	6.4	7.0	7.0	5.1	6.2		
		MCWBR	8.5	7.7	10.0	12.9	13.4	12.5	11.4	11.5	12.7	11.2	7.0	7.3		
(40) Clear-Sky Solar Irradiance	taub		0.272	0.307	0.354	0.393	0.403	0.400	0.423	0.405	0.365	0.343	0.305	0.273		
	taud		2.050	2.030	1.970	2.119	2.209	2.311	2.248	2.272	2.333	2.243	2.064	1.987		
	Ebn at Noon		615	735	786	820	837	845	814	803	788	696	556	498		
	Edh at Noon		65	99	134	135	131	120	126	115	94	80	65	55		
(44) All-Sky Solar Radiation	RadAvg		0.84	1.82	3.24	4.49	4.96	5.80	5.18	4.24	2.82	1.50	0.75	0.52		
	RadStd		0.09	0.15	0.24	0.26	0.56	0.55	0.36	0.37	0.33	0.15	0.10	0.06		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page